Sherif Morsi

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Summary

Concordia Computer Science student seeking AI/ML, software, and quant internships. Trained **PyTorch** models on 5G telemetry at Nokia; builds tested **Python** ML pipelines and leakage-safe forecasting and risk models with **AI agents** (Claude Code, Codex).

Education

Concordia University Montreal, QC, Canada
Bachelor of Computer Science *Expected May 2028*

- **Availability:** Winter 2027 (Jan to Apr 2027) for 4, 8, 12, or 16 months
- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Linear Algebra, Multivariable Calculus

Experience

AI & Machine Learning Intern Aug 2026 - Sep 2026
Nokia *Cairo, Egypt*

- Trained and evaluated Python and **PyTorch** models on wireless network telemetry to forecast 5G **energy demand**
- Developed **data and ML pipelines** for 5G energy telemetry, to support automated network management and optimization
- Presented findings on wireless networking and 5G-related energy challenges to cross-functional teams of data scientists and engineers; led the project proposal and requirements review

Credit Analyst Job Shadow Aug 2025 - Sep 2025
Qatar National Bank (QNB) *Cairo, Egypt*

- Shadowed credit analysts reviewing SME financial records as part of the bank's credit-review process
- Built **Python** utilities and **SQL** queries to preprocess and validate SME financial records, improving the data quality and reliability of credit-review data
- Tested and documented these scripts and fixes with the team so they stayed maintainable and easy to debug

Projects

Time-Series Foundation Model (Transformers, Forecasting) | *Python, PyTorch, pytest, GitHub Actions* | **GitHub** 2026

- Achieved **0.605**× **Seasonal Naive** zero-shot validation CRPS by pretraining a **10.1M-parameter** PyTorch **transformer**
- Validated **443,258 series** across **77 datasets** to keep evaluation benchmarks leakage-free, excluding **801** flagged series
- Researched and implemented the transformer and its training code in plain PyTorch, training in fp16 on a single T4 GPU
- Verified the ML pipelines with **138 automated tests** running in GitHub Actions CI

ATLAS (Deep Learning, Graph Models) | *Python, PyTorch, FastAPI, DuckDB, Playwright* | **GitHub** 2026

- Built Parquet/SQLite data pipelines for **42,900 World Bank records** with validation of normalized data and provenance
- Compared **12** PyTorch GRU and **graph forecasting** checkpoints on held-out baselines with reproducible, hash-verified replay
- Validated the **FastAPI**/DuckDB services with **44 backend tests**, **27 Playwright UI tests**, and cross-platform CI

World Shock Lab (Econometrics, Macro Shocks) | *Python, PyTorch, DuckDB, TypeScript* | **GitHub** | **Live Site** 2026

- Estimated Fed/BoC/OPEC surprise effects on **53 global assets**: a 10 bp Fed surprise moves the TSX −**1.10%**, BoC −**0.19%**
- Validated estimators by **simulation**, replicated **11 of 13** published findings, pre-registered an ML test on **168** announcements
- Shipped a React/TypeScript scenario tool (under **3 s** on a throttled phone) whose tests fail on any number not in the results

Certifications & Affiliations

Machine Learning with Python (**IBM**) | Claude Code 101 (**Anthropic**) | Claude 101 (**Anthropic**)
Quantitative Research and Competitions Club (QUARCC) | *Member, Concordia University*: Exploring machine learning concepts, statistical modeling, and quantitative research through data-driven problem-solving

Technical Skills

Languages: **Python**, **SQL**, C++, **Java**, C, R, JavaScript, TypeScript, Bash, MATLAB
Machine Learning: **PyTorch**, TensorFlow, **scikit-learn**, Deep Learning, Supervised/Unsupervised Learning, Regression, **Transformers**, **Time-Series Forecasting**
ML Methods: **Feature Engineering**, **Cross-Validation**, Transfer Learning, Computer Vision, NLP, **Model Evaluation**, Mixed Precision
AI / LLMs: **LLMs**, **AI Agents**, **Prompt Engineering**, Claude Code, **Cursor**, Codex, Hugging Face Transformers
Quant / Risk: **Econometrics**, Event Studies, Monte Carlo, Probabilistic Forecasting, **CVaR**, **Portfolio Optimization**, Walk-Forward Validation
Data / Libraries: **Pandas**, **NumPy**, SciPy, Matplotlib, OpenCV, **FastAPI**, **DuckDB**, **SQLite**, Parquet, cvxpy, statsmodels
Developer Tools: **Git**, **GitHub Actions (CI/CD)**, **pytest**, Playwright, Jupyter Notebook, Google Colab, VS Code, PyCharm
Office / BI: Excel (Advanced), Power BI, PowerPoint, Microsoft 365